

Solution Requirements

Date	23/09/2026
Team ID	SWTID-2026-3089
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Institution	Arignar Anna College Code: MSU 302
Team Members	Aldhaf F (Leader), Bhuvanasri S, Dhanisha Muhi N S, Divya Lakshmi K, Dharshini P
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform. Functional Requirements (FR) define specific behaviors, functions, or features of the system. Non-Functional Requirements (NFR) specify the operational criteria and performance standards.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Secure Single Sign-On (SSO) and LDAP integration within ServiceNow instance to authenticate users, approvers, and hardware engineers.	High
2	Authorization levels	Role-Based Access Control (RBAC): End users can order standard laptops; Managers have approval rights; Flow Designer executes with System User privileges; Hardware group members can manage Catalog Tasks.	High
3	External interfaces	Seamless internal interface binding between ServiceNow Service Catalog (`sc_cat_item`), Flow Designer engine, and Task Dispatch subsystem (`sc_task`).	High

4	Transactions processing	Automated transaction processing on `sc_req_item`: Upon approval event, Flow Designer automatically creates a Catalog Task, links it to RITM, populates Short Description as 'Laptop need to Configured', and assigns group to 'Hardware'.	High
5	Reporting	Real-time visual dashboard reporting displaying standard laptop order volume, average configuration lead time, pending approvals, and active task distribution across Hardware technicians.	Medium
6	Business rules & workflow	Automated flow triggered by 'Service Catalog', associated with 'Standard Laptop' catalog item in Maintain Items under Process Engine, replacing legacy manual workflows.	High
7	Compliance to regulations	Comprehensive audit logging on all approval changes and task modifications in compliance with ITIL v4 Service Management and corporate asset governance guidelines.	Medium
8	Data validation	Automated form validation ensuring all required recipient attributes, cost center, and laptop configuration specs are validated before task generation.	Medium

Step 2: Non-Functional Requirements (NFR)

NFR Category	Requirement Description	Target Metric / Criteria
Performance & Speed	Flow Designer trigger and Catalog Task creation must execute immediately upon manager approval without human lag.	Task generated in < 2 seconds post-approval.
Scalability	System must handle organizational scaling, supporting enterprise-wide concurrent catalog orders across multiple departments.	Supports 5,000+ simultaneous requests without latency.
Security & Data Privacy	Data encrypted in transit (TLS 1.3) and at rest (AES-256). Strict ServiceNow Access Control Lists (ACLs) restrict task modifications to Hardware team.	100% adherence to corporate IT security standards.

Reliability & Availability	Continuous operation ensuring procurement workflows execute seamlessly 24/7 without dropped triggers or stuck records.	99.9% platform availability (ServiceNow SLA).
Usability & Accessibility	Clean, user-friendly Service Catalog interface allowing any employee to order a standard laptop in under 2 minutes.	WCAG 2.1 AA compliant; < 3 clicks to checkout.
Maintainability	Low-code / no-code Flow Designer workflow in Global scope, enabling easy maintenance and extension to desktop and mobile devices.	Zero custom Java/C++ maintenance overhead.